AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956
COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I142*

Team Name: *QuadriX*

Domain: *1*

Amrita Vishwa Vidyapeetham, Coimbatore Campus

Date: 11th October 2026

Presentation Outline

1. Problem, Understanding & Gaps — slide 3
2. Our Proposed Solution — slide 4
3. Technical Approach — slide 5
4. Prototype: Extract & Validate — slide 6
5. Prototype: Government Officer — slide 7
6. Prototype: Marketplace — slide 8
7. Innovation & Uniqueness — slide 9
8. Feasibility, Scalability & Cost — slide 10
9. Use Cases — slide 11
10. Use Case Walkthrough — slide 12
11. Impacts, Benefits & SDGs — slide 13
12. Individual Member Contribution — slide 14

Problem, Understanding & Gaps

Problem Statement: A property is represented by many records instead of one decision-ready profile. Its evidence is distributed across deeds, revenue records, EC, mutation, survey and sometimes GIS and other records. Today, a user or officer often has to manually connect those pieces then verify it and also people cant purchase a desired plot online based on verified documents data instead they depend on ADS.

We aim to solve: Turn fragmented property evidence into one connected, explainable and actionable intelligence layer; extract the data summarize into understandable forms and then validate it. Then use this verified data to give property suggestion under marketplace.

Problem Understanding

Who experiences this problem?

- Citizens / buyers verifying a property — citizens often do not know which missing evidence matters most and wont understand the doc language easily.
- Government officers (Taluk/department) — they should not have to repeat every comparison manually.
- Buyers and sellers on property portals — normal property portals optimize results based on ADS, not evidence-aware verification.

What causes the problem?

- A property is represented by many records instead of one decision-ready profile.
- The same owner can appear with initials, spelling differences or transliteration.
- Area, survey/subdivision, boundaries and ownership may differ across records.
- Old/scanned/handwritten records are difficult to search and compare manually and interpret later.
- EC contains transaction information that must be interpreted in context.
- Mutation links transactions to revenue/current records but may need reconciliation with other evidence.
- Survey/FMB/cadastral/GIS evidence adds a spatial dimension but may live separately.
- Platforms sell plots based on AD system rather than verified details of seller.

Existing Gaps & Challenges

Gap 1 — No end-to-end evidence system: The research survey identifies a major India-specific gap in an end-to-end system connecting Patta/RoR, deeds, mutation, EC, registration and cadastral/GIS with title/encumbrance/risk-oriented analysis. Existing tools solve one piece: OCR, land-record digitization, GIS, EC/mutation analysis, a government dashboard or a property marketplace — not the connected chain.

Gap 2 — Discovery without evidence: Normal property portals optimize discovery, not evidence-aware verification. The survey also identifies buyer-property matching using title confidence, encumbrance, mutation completeness and parcel certainty as an underdeveloped India-specific area.

Gap 3 — Scores without explanation: A simple red risk score does not tell a citizen or officer what to do. A score without evidence and explanation is not actionable. Extraction confidence, evidence strength and verification status are usually collapsed into one number — OCR can be 99% confident that the document says 2.50 acres while the property still has a real 0.15-acre conflict.

Main challenge that remains unsolved: Reconciling heterogeneous records into one trustworthy, explainable profile — when different systems can use different identifiers, formats, update cycles and standards, and live government integrations require authorized APIs/data-sharing mechanisms — *while keeping uncertainty visible and leaving consequential legal decisions to authorized humans/officials.*

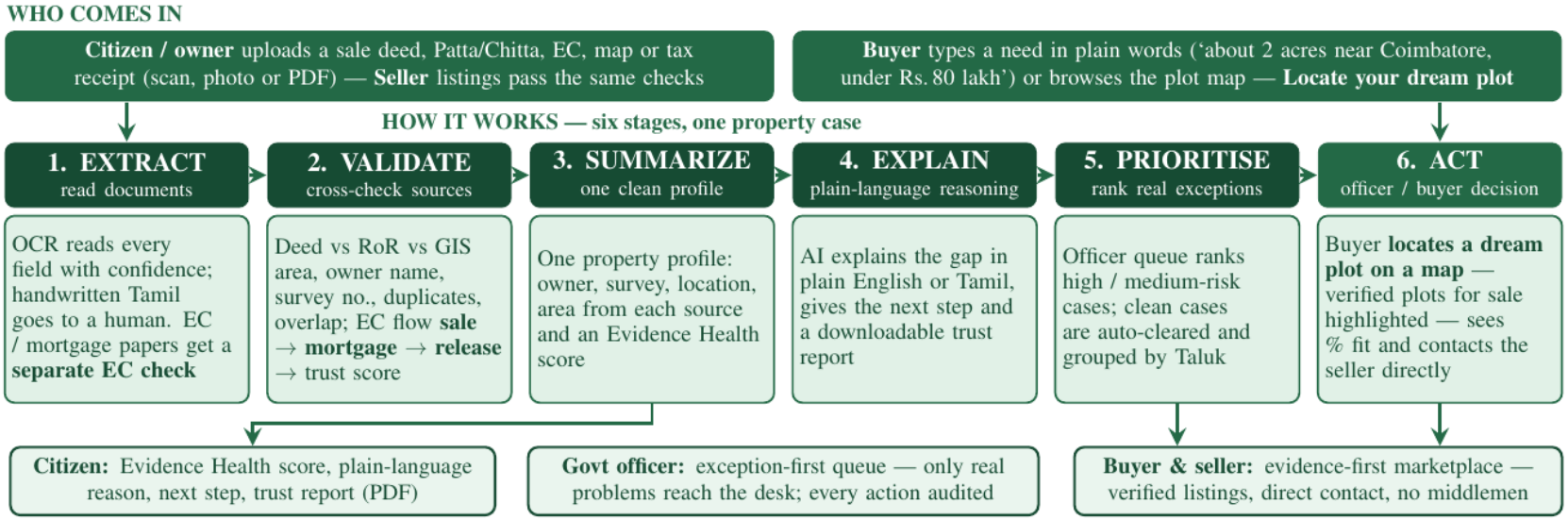
Problem Understanding

How does it affect users or stakeholders?

- No decision-ready profile: users must manually connect deeds, Patta/RoR, EC, mutation and survey records.
- A simple red risk score does not tell a citizen or officer what to do.
- Officers repeat comparisons case by case instead of focusing on exceptions.
- Buyers see '2.5 acres – Rs. 40 lakh – Contact Seller', not the evidence behind it.

Our Proposed Solution

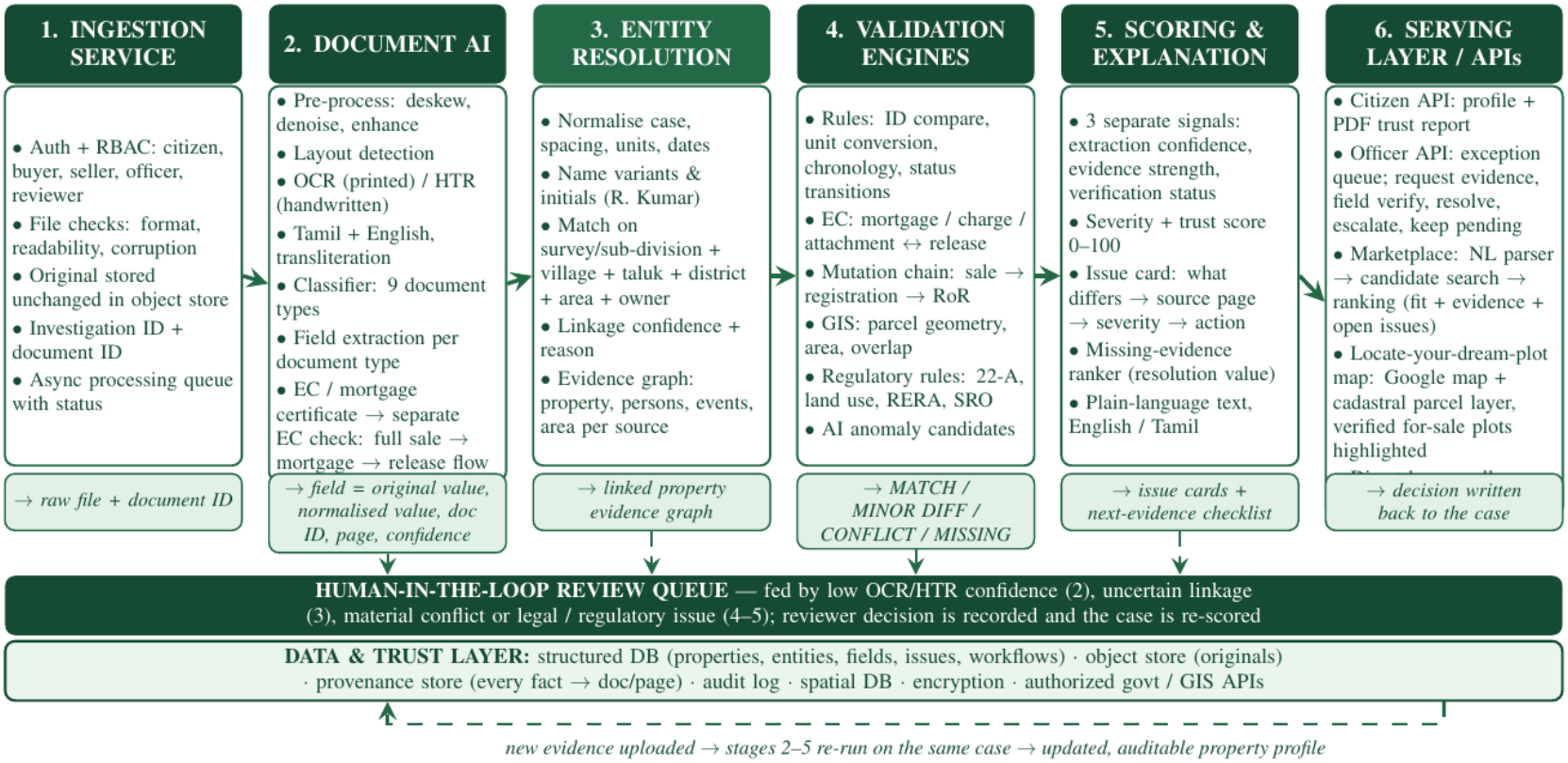
- **Core idea or product:** LAND LENS is an evidence-first property intelligence layer. A user uploads whatever property documents are available. The system identifies them, reads them, extracts structured facts, connects records belonging to the same property/person, compares the evidence, checks EC/mutation/GIS and applicable regulatory questions, explains inconsistencies, recommends the next evidence to obtain, and routes important cases to human/official review.
- **How the solution works at a high level — the six stages of the working prototype:**



- **Main value delivered to users:** The individual technologies are not claimed as new. The system-level combination turns fragmented property evidence into one connected, explainable and actionable intelligence layer — citizens get a simple Property Intelligence report, government officers get an exception-first dashboard, and buyers/sellers get an evidence-first marketplace.

Technical Approach

SYSTEM ARCHITECTURE — DATA FLOW FROM UPLOADED FILE TO DECISION

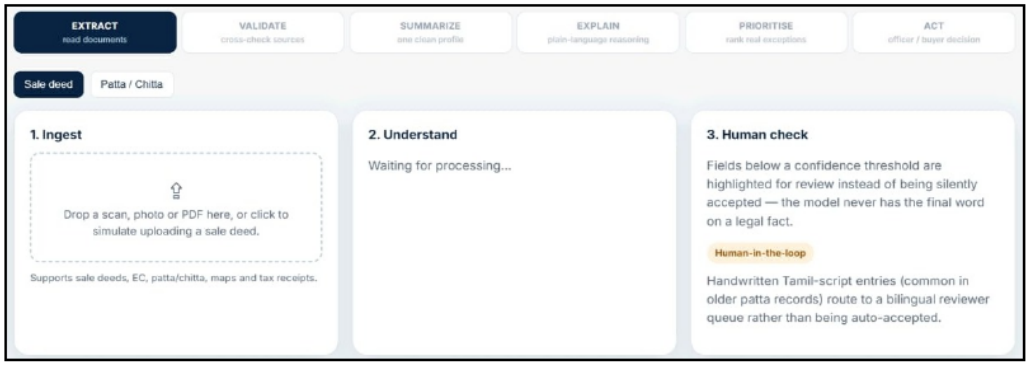


Tech Stack

Frontend: Citizen, seller, buyer and officer web portals; document viewer with source page & confidence; GIS parcel map.
Database: Structured DB + object storage for originals + provenance metadata; optional spatial (GIS) DB.

Backend: Case & ingestion service, evidence-graph service, rule engine (validation, EC, mutation, regulatory), matching/ranking, audit & RBAC.
AI/ML, APIs: OCR/HTR, document classification, entity extraction & resolution, anomaly detection, NL requirement parsing; authorized govt/GIS APIs.

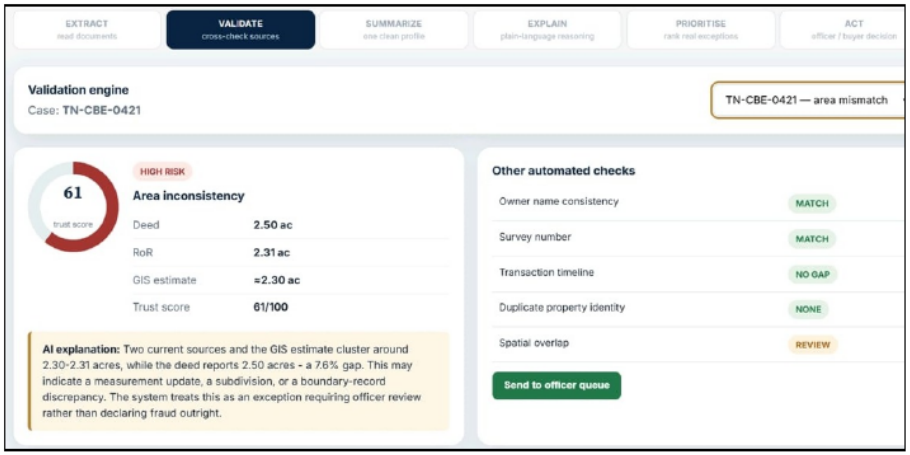
Technical Approach — Prototype: Extract & Validate



1 Extract

Stage: read documents

Upload a scan, photo or PDF; fields are read with confidence and low-confidence or handwritten Tamil entries go to a human reviewer.



2 Validate

Stage: cross-check sources

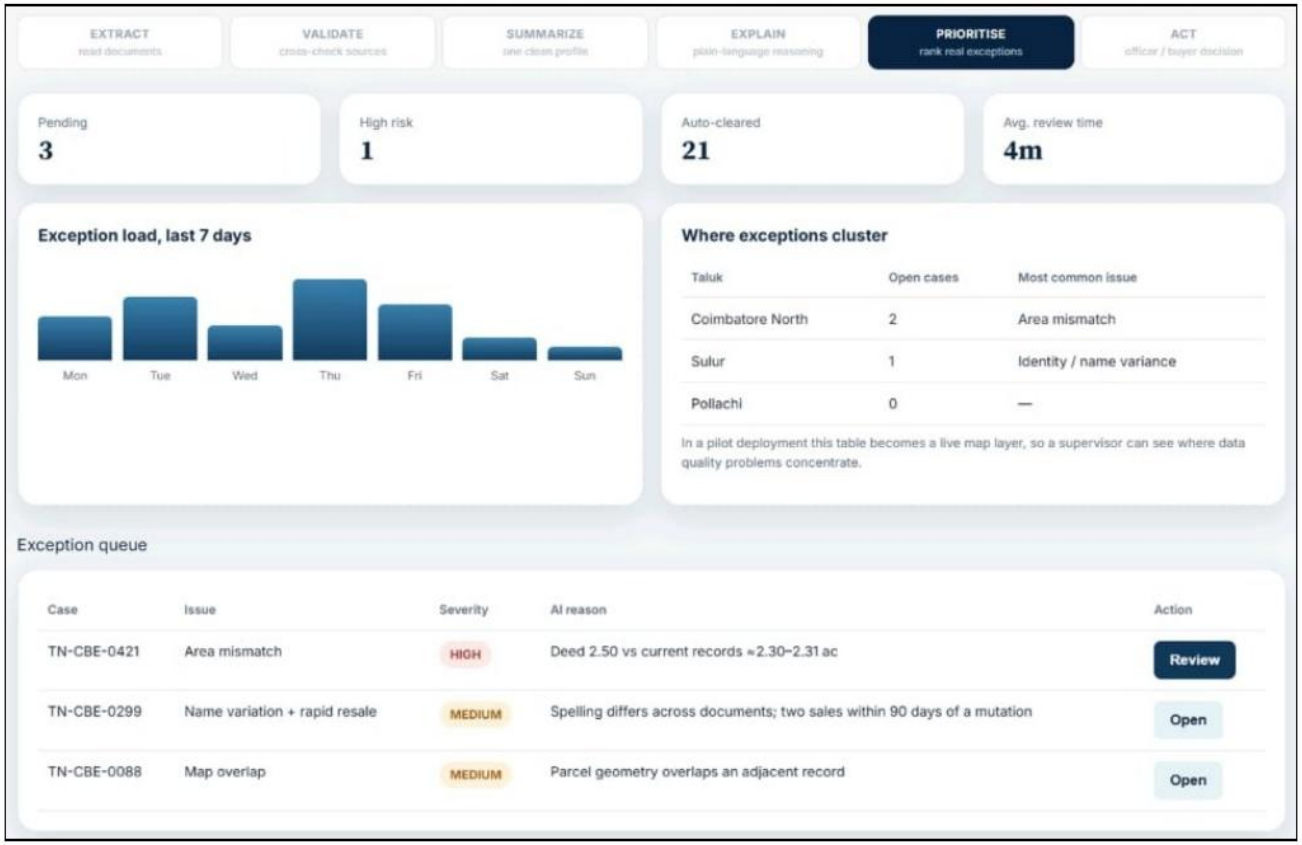
Deed 2.50 ac vs RoR 2.31 ac vs GIS ≈2.30 ac is flagged as a 7.6% gap and sent to the officer queue, not called fraud.

Separate EC check when an EC or mortgage certificate is uploaded, its full flow is shown:

2016 — Sale deed registered → **2021 — Mortgage (charge) created** → **2024 — Release (discharge) found** → **Result — no open charge; source page & confidence shown**

If a mortgage has no matching release, the check stays **open** and 'get the release / discharge document' is added to the next-evidence list.

Technical Approach — Prototype: Government Officer

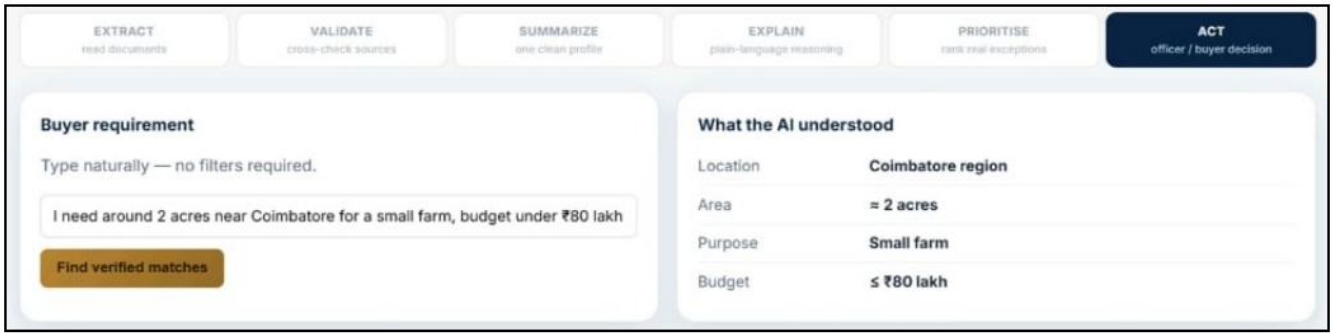


3 Government Officer Console

Stage: Prioritise — rank real exceptions

Pending, high-risk and auto-cleared counts, Taluk-wise clusters and a severity-ranked exception queue — the officer opens only the cases that need attention.

Technical Approach — Prototype: Marketplace



4 Buyer need AI reads location, area, purpose and budget from a plain-language need.



5 Verified matches Ranked by % fit and evidence; buyer contacts a verified seller directly.

6 Locate your dream plot — map view

A digital map (Google map base + cadastral parcel layer) shows the buyer's area; plots available for sale are highlighted — ■ verified, ■ under review. Tapping a plot opens its evidence profile, % fit and 'request direct contact'.

Innovation & Uniqueness

What is new or different?

- **One profile across 9 document types** — Sale Deed, Patta/RoR, EC, Mutation, Registration, Tax, Survey/FMB, Cadastral/GIS and others — instead of a single-document OCR or search tool.
- **Conflicts are shown, not overwritten:** deed 2.50 ac vs Survey/FMB 2.35 ac is reported as a 0.15-ac conflict with the source page.
- **Three separate signals, not one red score:** extraction confidence, evidence strength and verification status — OCR can be 99% sure the deed says 2.50 ac while the 0.15-ac conflict is still real.
- **Tells the user what to get next:** latest Survey/FMB, mortgage release/discharge or latest Mutation/RoR, ranked by how much each resolves.
- **Marketplace ranks by evidence,** not only area, price and location (prototype: 92% / 89% / 76% fit with verification status).

Why is this approach suited to the problem?

- **Legacy records:** the research survey names pre-processing + OCR/HTR as the base method for scanned and handwritten land records — our Extract layer, with handwritten Tamil routed to a bilingual reviewer.
- **Fragmented identities:** entity extraction, normalisation and record linkage are core to an end-to-end land-record workflow — matches Ramesh Kumar / R. Kumar / Ramesh K. only when property evidence agrees.
- **India-specific gap:** no end-to-end system links Patta/RoR, deeds, mutation, EC, registration and cadastral/GIS with title/encumbrance/risk analysis — LAND LENS targets exactly this.
- **Second gap:** buyer–property matching using title confidence, encumbrance, mutation completeness and parcel certainty is underdeveloped in India — our evidence-first marketplace.
- **Local fit:** Taluk-level officer queue and Tamil Nadu Section 22-A restriction checks.

What is the key differentiator?

Area	Typical tools today	LAND LENS
Property portals	'2.5 acres – Rs. 40 lakh – Contact Seller'; ranking driven by ads	Verified vs unresolved shown on every listing; ranked by fit + evidence strength + open issues
Digitisation / OCR	Text output from one document	Structured fields with page & confidence, linked across all records of the property
Risk checks	One score, no reason	Issue card: what differs → source → severity → next evidence (prototype: 2.50 vs 2.31 vs ≈2.30 ac = 7.6% gap)
Government	Officer compares every document by hand	Exception-first queue (prototype: 21 auto-cleared, 3 pending, 1 high-risk) with full audit trail

Non-Negotiable Principles

- Never fabricate government access or live data.
- Never convert AI confidence into legal certification.
- Never silently overwrite conflicting source values.
- Always preserve provenance.
- Use human/official review for consequential decisions.
- Protect sensitive documents.
- Distinguish source values, normalized values, AI inferences and official outcomes.
- Use exception-first government workflows.
- Keep marketplace and verification connected through the same evidence graph.
- Treat uncertainty as an explicit output.

Feasibility, Scalability & Cost

Feasibility

Prototype: Use synthetic/representative property records to demonstrate the workflow.

Production: Combine user/seller uploads with authorized government APIs, approved data-sharing arrangements and permitted GIS/registration sources.

Build: Proven building blocks (OCR/HTR, classification, entity extraction, record linkage, rules engine, GIS connector) assembled as services; AI + deterministic rules + human-in-loop.

Scalability

- One evidence graph, many interfaces: citizen, government and marketplace workflows all use the same evidence foundation.
- Taluk-wide exception-first dashboard: automated screening → identify exceptions → prioritize → officer investigates selected cases.
- New evidence can be uploaded and the same case revalidated.
- Jurisdiction-specific regulatory checks (e.g., Tamil Nadu Section 22-A) added as rules per state.

Cost for Product (Optional)

- **Development or hardware cost:** Prototype built on open-source OCR/HTR, NLP and GIS libraries with synthetic/representative data; no dedicated hardware. Indicative: Rs. 0 – Rs. 50,000 (cloud credits, test scanning, domain/SSL).
- **Hosting or maintenance cost:** Pilot for one Taluk on a managed cloud — application server, OCR worker, relational + object storage, optional spatial database. Indicative: Rs. 15,000 – Rs. 40,000 per month.
- **Estimated total:** Indicative 6-month pilot: Rs. 1.0 – Rs. 3.0 lakh (INR), excluding team effort and any government data-access/integration charges.

Basis of estimate: indicative team assumptions for a prototype/pilot, not vendor quotations. Production cost depends on scan volume, OCR/HTR model choice, GIS licensing and the authorized government integration route — to be validated before submission.

Risks: main constraints and possible mitigations (what happens when data is wrong?)

- If OCR is uncertain → retain confidence and request review.
- If two records disagree → preserve both and flag conflict.
- If property linkage is uncertain → do not force the match.
- If GIS is unavailable → mark unavailable.
- If a regulatory rule cannot be reliably applied → show verification required rather than inventing an answer.
- If government data is unavailable → do not claim it was checked.
- If the user lacks documents → identify the most useful next evidence.

Deployment challenge: Different systems can use different identifiers, formats, update cycles and standards. **Correct product behavior:** Show source and availability/freshness where possible and clearly indicate when a check could not be performed.

Do not claim: Unrestricted access to Tamil Nadu government databases.

Security, Privacy, Provenance & Audit

- Role-based access for citizen, buyer, seller, officer and authorized reviewer.
- Encryption for sensitive documents/data.
- Sensitive original records are not publicly exposed through marketplace listings.
- Original source values remain distinguishable from normalized/AI-derived values.
- Every important result keeps document/page provenance.
- Officer actions and major status changes are auditable.
- Synthetic/representative prototype data is clearly labeled.
- Live government integrations require authorized APIs/data-sharing mechanisms.

Use Cases

ONE PROPERTY → ONE EVIDENCE GRAPH → THREE DECISION INTERFACES

The project is not three separate modules. All three use cases are powered by the same Property Intelligence Engine. The same property evidence is understood, connected, validated and continuously updated, then presented differently to a citizen, a government officer and a buyer/seller.

USE CASE 1 · Verify a Property

Citizen / Property Owner · Understand and validate property evidence

Goal: Let a citizen understand the available evidence for a property without requiring them to manually compare multiple documents.

User journey

1. The citizen clicks 'Verify Property' and receives investigation ID PROP-2026-001.
2. They upload whatever documents they have; they do not need to know the technical document type.
3. AI identifies the documents and extracts owners, survey numbers, areas, dates, transactions and encumbrance information.
4. The system links records that appear to describe the same property.
5. A unified Property Intelligence Profile is created with source/page/confidence information.
6. Cross-record validation compares deed, Patta/RoR, mutation, EC and survey/GIS evidence where available.
7. The system identifies the 0.15-acre discrepancy and explains exactly which evidence creates it.
8. The Missing Evidence Engine recommends the latest official survey/FMB as the next useful evidence.
9. The citizen sees Evidence Health, issues, missing evidence and next actions in simple language.

What the user sees: Documents → Unified Property Profile → 2.50 vs 2.35 acre conflict → explanation → next evidence.

Why this use case matters: This is the core intelligence engine. It demonstrates that the product does more than store or digitize documents: it connects evidence and reasons over inconsistencies.

USE CASE 2 · Exception-First Verification

Government Officer · Prioritize and investigate problematic cases

Goal: Help officers focus their time on cases that actually require investigation.

User journey

1. Multiple property cases are processed through the same intelligence and validation pipeline.
2. The system automatically identifies conflicts, missing evidence, unresolved encumbrances and regulatory verification needs.
3. Cases are prioritized into an exception-first queue instead of requiring the officer to manually start from page one for every property.
4. The officer opens PROP-2026-001 and sees the exact 0.15-acre discrepancy, supporting evidence and source/page.
5. The system explains why the case was prioritized and suggests the next evidence or verification action.
6. The officer can Request Evidence, Field Verify, Resolve, Escalate or Keep Pending.
7. Every decision/action is recorded in the audit trail.

What the user sees: All cases → AI detects exceptions → officer queue → PROP-2026-001 → evidence-backed investigation → official action.

Why this use case matters: The government workflow proves that the same intelligence can reduce repetitive comparison and help officers prioritize exception cases while keeping humans responsible for official decisions.

USE CASE 3 · Evidence-First Marketplace

Buyer + Seller · Discover and match properties using requirements + evidence

Goal: Turn property intelligence into a more evidence-aware discovery and transaction experience.

Seller journey

1. Seller enters basic property details and uploads available evidence.
2. The same document intelligence and validation pipeline creates an evidence-backed property profile.
3. The listing shows evidence strength, available verification status and unresolved issues instead of hiding them.
4. Sensitive documents remain private and can be shared through controlled evidence requests.

Buyer journey

1. Buyer describes a requirement in natural language, such as: '2–3 acres near Coimbatore, within Rs. 40 lakh, suitable for agriculture, with strong documentation.'
2. The system converts the request into structured requirements: location, area, budget, suitability and evidence preference.
3. Listings are matched using both property characteristics and evidence strength.
4. The buyer sees match percentage, evidence status, unresolved issues and the reasons for the match.
5. The buyer can View Evidence, Request Evidence, Save Property or Contact Seller.

What the user sees: Evidence-backed listing → buyer requirement → requirement + evidence-aware matching → transparent property result → buyer/seller connection.

Why this use case matters: The marketplace is not an unrelated add-on. It is a downstream consumer of the same Property Intelligence Profile, so discovery can consider both what the buyer wants and how much evidence is available.

Use Case Walkthrough — PROP-2026-001

Common scenario used across all three cases

Use the same fictional property investigation, PROP-2026-001, throughout the demonstration. This makes the project feel like one real system instead of three disconnected demonstrations.

Property Evidence	Example
Sale Deed	**2.50 acres**
Patta / RoR	**2.50 acres**
Survey / FMB	**2.35 acres**
EC	Transaction / encumbrance history available
Mutation	Ownership transition evidence

The system therefore detects a **0.15-acre difference** between the Survey/FMB evidence and the other records. The conflict is preserved and explained rather than silently overwritten.

Important boundaries — to avoid user confusion

- The system does not automatically declare a property legally valid or invalidate legal title.
- It distinguishes extraction confidence, evidence strength and verification status.
- Conflicts are preserved with their supporting sources; the AI does not silently choose one record as truth.
- GIS or government data is used only when authorized/available; unavailable evidence is clearly marked.
- Marketplace listings show unresolved issues transparently; the platform should not claim '100% legally verified' solely from AI output.

Complete product flow

Stage	What happens	What the user gets
1	**Evidence enters the system**	Citizen, seller or authorized source provides available property documents.
2	**AI understands documents**	Document classification + OCR/HTR + field extraction + confidence + provenance.
3	**Records are connected**	Entity/property resolution links documents referring to the same property.
4	**Unified Property Intelligence Profile**	All available evidence is organized around one property identity.
5	**Evidence is cross-validated**	Ownership, area, survey, mutation, EC, transactions and GIS/survey evidence are compared.
6	**Issues are explained**	Conflicts, missing evidence and applicable regulatory checks are surfaced with sources.
7	**Human/official verification**	The system recommends the next evidence or action; it does not make a legal title decision.
8	**Three outputs**	Citizen intelligence • Government exception review • Evidence-first marketplace.
9	**New evidence returns to the loop**	Updated evidence can trigger revalidation of the same property.

The critical connection



Impacts, Benefits & SDGs

Benefit to the target users

- **Families & landowners:** understand their own land papers in simple English or Tamil and know what is missing — before they sign or pay.
- **Buyers:** find checked plots on a map, avoid fake or disputed land and deal directly with the real owner — no middlemen, no ad-driven listings.
- **Honest sellers:** a verified listing earns trust and sells faster.
- **Government officers:** time goes to real problem cases; every decision is open and recorded.

Social impact

- **Trust in land deals:** the same facts are visible to buyer, seller and officer.
- **Protection from fraud:** first-time buyers, farmers and elderly owners see red flags early.
- **Fewer land disputes:** mismatches are caught before money changes hands.
- **Inclusion:** plain language, local language — no lawyer needed just to understand a document.

Expected measurable outcome (to be measured in a pilot Taluk)

- Time for a citizen to understand a property's papers: from repeated office visits to one online report.
- Share of marketplace listings verified before sale.
- Officer hours saved per case; cases cleared without manual comparison.
- Fraud / dispute cases flagged before registration.

SDG	Target	Goal	How LAND LENS helps
SDG 1	1.4	No Poverty	Families keep secure ownership of their land — often their biggest asset.
SDG 10	10.2	Reduced Inequalities	Every citizen, not only experts, can understand land records in their own language.
SDG 11	11.3	Sustainable Cities & Communities	Buyers choose safe, legal plots on a map, supporting planned growth.
SDG 16	16.6	Peace, Justice & Strong Institutions	Fewer land disputes and fraud; open, recorded government decisions.

Individual Member Contribution

Team Member	Role	Responsibility in Execution
Kirthi Madala	AI / Document Intelligence	OCR/HTR, multilingual document extraction, field detection and confidence scoring
Monish Karthik S	Validation & Land Intelligence	Cross-record validation, EC/mutation/title checks, anomaly detection and evidence scoring
K.A. Tirunavukkarasu	Full-Stack & Platform	Web portal, backend APIs, database, authentication and officer/citizen workflows
Surya P	Marketplace & Product	Evidence-first marketplace, buyer–property matching, UI/UX and user workflow testing

Yukti ID

IR2026-1051102